

Nanya Technology

2Q26 a slight beat; rising server mix coupled with favorable LTAs to bode well for mid-long term earnings; reiterate OW

NYT's 2Q results were a slight beat vs. JPMe/consensus thanks to sustained pricing momentum (DRAM ASPs up >60%) alongside a flattish bit shipment trend. Management raised its 2026E bit growth guidance to high-teens % Y/Y (from teens % in 1Q26), citing lower inventory levels vs historical trends, and reiterated its 2026 capex guidance of NT\$52bn, pointing to severe industry-wide shortages (S-D shortage continuing into 2028E, consistent with [Micron commentary](#)). We expect both DDR5 and DDR4 pricing to trend higher in 2H26E as the demand outstripping supply environment continues. We found NYT's server sales mix exposure increase as a positive development and its product diversification into LPDDR5 (likely for server applications in our view) and other line-ups. Mid-to-long term capex plan of US\$16bn for 45k wspm capacity sounded higher than our initial expectation; however, we view overall capital intensity under control and expect an increase in R&D spending to support NYT's product diversification initiatives amidst its flexibility to move around between DDR5 and DDR4. We expect volatile NYT share sentiment to continue in the near-term until CSP's capex guidance upward revision is confirmed in July earnings result call season. Risk-reward profile remains favorable on a medium-term horizon and we recommend investors accumulate the stock on our bullish top-down memory up-cycle view ("higher for longer"). Next catalysts are: 1) CSP's hardware capex spending and AI monetization progress update, 2) Asian memory peers' outlook commentary, 3) DDR4/DDR5 monthly pricing update, and 4) China memory competitor's mid-term business plan update post IPO.

- 2Q26 result flash and 2H26E update.** NYT reported OP of NT\$60.8bn on sales of NT\$82.5bn (OPM: 74%), driven by flat bit shipment growth (JPMe: -MSD%) and high-60% ASP growth (JPMe: 70%+ growth), which together delivered strong operating leverage. Before new capacity installation translates into a production bit growth (post flattish H/H bit shipment trend in 1H26, implied 2H26E bit shipment is likely to be down H/H), we expect pricing to be the key earnings driver in 2H26E and expect blended ASP increase of +20%/+11% each respectively in 3Q26E/4Q26E. Of note, DDR4 pricing momentum remained stronger vs. DDR5 in 2Q26, and we expect such a trend to continue in the short-term. Interestingly, AI infrastructure and server applications accounted for >20% of NYT's sales and we found the progress constructive given the firm's historical server compute sales exposure has ranged between 10-15% in the past. Including eSSD (indirect server exposure), we believe actual server compute exposure to be higher. We continue to view NYT's attempt to enter the server ARM CPU supply chain via the LPDDR5X stacked solution product offering as the next major driver and expect a rising server exposure to bode well for NYT's earnings profile in the near to mid term as highlighted in our recently published OW upgrade note ([link](#)).
- Various LTAs to help dampen earnings cyclicality.** Management noted that, amid supply tightness expected to persist for multiple quarters, customers are increasingly entering into multi-year LTAs, which improve planning visibility

Overweight

2408.TW, 2408 TT
Price (09 Jul 26):NT\$435.50
Price Target (Jun-27):NT\$710.00

Technology - Semiconductors

Jay Kwon ^{AC}

(82-2) 758-5725
jay.h.kwon@jpmorgan.com
J.P. Morgan Securities (Far East) Limited, Seoul Branch

Sangsik Lee

(82-2) 758 5146
sangsik.lee@jpmorgan.com
J.P. Morgan Securities (Far East) Limited, Seoul Branch

Neelay Y Kamath

(91-22) 6157 3764
neelay.kamath@jpmchase.com
J.P. Morgan India Private Limited

See page 8 for analyst certification and important disclosures, including non-US analyst disclosures.

and can help underpin pricing. In parallel, incremental “greenfield” capacity additions appear to be tied to multi-year LTAs extending beyond 2028, giving NYT a clearer line of sight into both near-term requirements and longer-term demand commitments. The company also emphasized that LTAs vary in structure: some lock in both volume and price for the contract term (with renewal upon expiry), while others commit to volumes but set pricing based on prevailing market conditions, typically reset on a quarterly cadence. Overall, we expect Nanya Technology to continue expanding its use of LTAs, which should help dampen earnings cyclicality relative to what the memory industry has historically experienced over the past two decades.

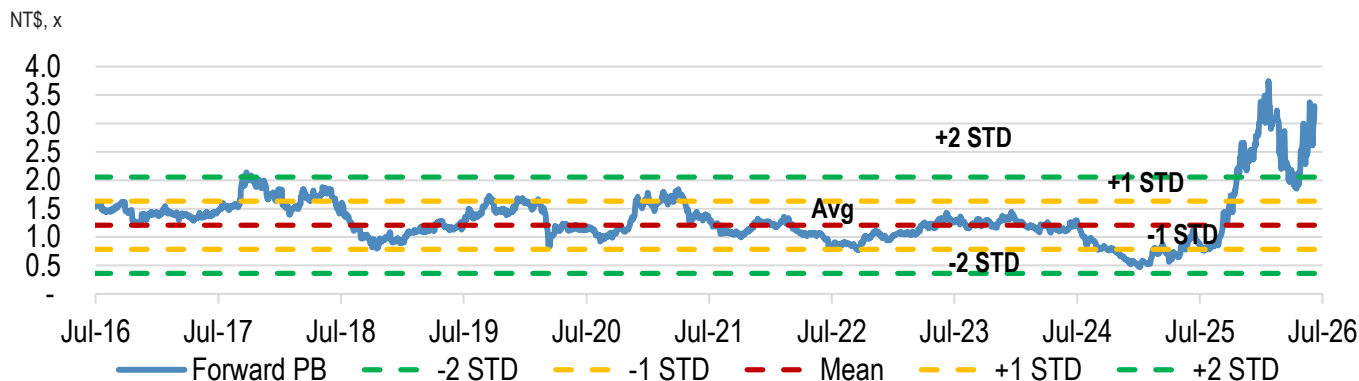
- **Capex update and implications to 2027E-2028E bit production.** The company reiterated its plan to spend NT\$52bn (in-line with JPMe) on capex in 2026 (1H run-rate of NT\$6.9bn, highlighting an unprecedented 2H spend of ~NT\$45bn) driven by the severe memory shortage being experienced in the industry. While the first phase of the new fab capacity expansion is planned to reach 30k wspm by 2028, peak output is expected to reach 45k wspm, with total capex estimated at NT\$480 billion, with construction cost near US\$3bn in total. Compared to our NT\$172bn estimate for 2026E-28E, this is significantly higher. We would like to highlight that such higher capex spending is a strong supply debottlenecking signal from NYT management amidst growing LTA discussions which should support the memory cycle moving with a lot more mid-to-long term stability. As a result of higher capex, we foresee a meaningful bit increase from 2028E onwards (JPMe: 2027E/2028E bit supply growth assumption at: +5%/+27%).
- **Stock implications.** NYT stock gained 114% (vs. TWSE +39%) in the 2Q26 driven by strong DRAM pricing momentum as well as positive EPS upgrades. However, following strong MU results in late June’26 ([link](#)), memory share sentiment has been weak with NYT shares declining 17% from its peak (vs. Global memory peers -14% or SOX/TWSE -4%/-1%) due mainly to growing AI capex, continuity concerns post Meta headline ([link](#)). However, following a strong monthly update and Meta’s 2x capacity build plan release ([link](#)), the stock has started to bounce back and is up 10% from the recent bottom. Until CSP’s hardware capex guidance upward revisions signal becomes evident, we expect volatile range-bound share sentiment could continue. From an overall impression standpoint, we came back with more positives (near-term price strength, product mix improvement toward servers, solid LTA progress, and top-down S-D shortage view extending into 2028E) vs. negatives (substantially higher capex spending into 2028E) from the NYT results, and view the stock as risk-reward favorable from a midterm horizon. Maintain OW view on the stock and recommend investors accumulate.

Table 1: NYT 2Q earnings review table

(NT\$ mn)	2Q25	1Q26	2Q26 Actuals	Q/Q (%)	Y/Y (%)	JPM	Diff. (%)	BBG	Diff. (%)
Sales	10,526	49,087	82,549	68.2%	684.2%	81,279	2%	81,185	2%
Gross profit	(2,165)	33,316	65,619	97.0%	na	63,158	4%	64,061	2%
Gross Margin	-21%	68%	79%	1162 bps	10006 bps	78%	-179 bps	79%	58 bps
Operating profit	(4,501)	30,111	60,826	102.0%	na	59,244	3%	60,159	1%
Operating Margin	-43%	61%	74%	1234 bps	11645 bps	73%	-80 bps	72%	195 bps
Net Income	(4,102)	26,059	50,192	92.6%	na	49,454	1%	48,930	3%
Net Income Margin (%)	-39%	53%	61%	771 bps	9977 bps	61%	4 bps	60%	53 bps
New Taiwan GAAP EPS	(1.32)	8.41	14.55	73%	na	14.33	1%	14.57	0%

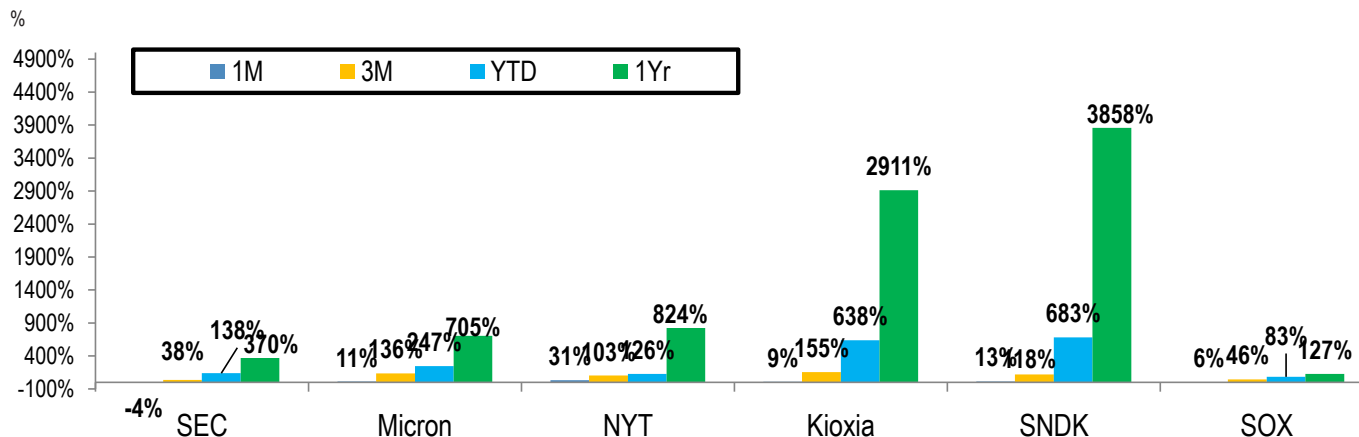
Source: Company data, Bloomberg Finance L.P., J.P. Morgan estimates.

Figure 1: NYT – FTM P/B band



Source: Bloomberg Finance L.P., Company data, J.P. Morgan estimates

Figure 2: Global memory makers' share price performance including SOX (Philadelphia Semiconductor index)



Source: Bloomberg Finance L.P. Note: Past performance is not an indicator of future results

Q&A highlights

Q. What is the pricing trend for DDR4 and DDR5 into 3Q26?

A. Pricing continues to move higher. Short-term agreements offer greater room for ASP upside, while long-term agreements provide better stability.

Q. What is the magnitude of further price increases?

A. No specific guidance. LTA pricing should remain relatively stable, while short-term pricing may continue to improve as contracts renew.

Q. How do the DRAM S/D dynamics look into 2028-2029 as peers add new capacity?

A. The S/D environment is not expected to change drastically by 2028E. Despite new capacity additions, AI/server demand and supply constraints across non-AI applications should keep the overall market tight.

Q. Was the key driver of 2Q26 earnings the 60%+ ASP increase despite flat bit shipments and limited cost changes?

A. Yes. Bit shipments were flat Q/Q and the cost structure was broadly stable, with limited cost reduction. The sharp earnings improvement was mainly driven by higher ASP.

Q. Does 60%+ Q/Q ASP growth imply most of the business is still based on monthly price negotiations despite prior LTA announcements?

A. LTAs are generally based on fair market pricing at the time of negotiation. LTAs do not prevent price upside in a rising market.

Q. Are LTAs designed to fix price, volume, or both?

A. LTA structures differ by customer. Some agreements fix volume, some include pricing commitments, and some include both. Pricing and volume terms are often reviewed with reference to market trends, typically on a quarterly basis.

Q. What is the outlook for 3Q/4Q ASP and operating results?

A. 3Q operating results are expected to further improve, with OPM already at a high level and likely to move higher. For 4Q, visibility depends on market conditions.

Q. Any impact from local Chinese DRAM makers' revenue or production increase?

A. There is no direct impact on NYT currently. The business remains stable globally, and customer/product overlap appears limited.

Q. What is the product revenue mix?

A. DDR4 remains the largest contributor. DDR4 and LPDDR4 together account for roughly 60-70% of revenue, while DDR3 and DDR5 each account for approximately 10%, with the remainder split across other products.

Q. Any color on the NT\$16 billion capex for 45K WSPM?

A. Construction costs are less than NT\$3bn and the rest is equipment including EUV. EUV adoption is expected around 2028E, potentially for 1C/1D/1E depending on equipment arrangement and technology timing.

Q. Are there any updates on customized DRAM/AI solutions and HBM development?

A. Customized DRAM projects are progressing with multiple customers. Projects are also ongoing on HBM-related extensions, including wafer-to-wafer bonding, with product timing dependent on customer project schedules.

Q. What is the outlook for non-AI products into 2028?

A. High-end non-AI products are performing better than low-end products. NYT will continue to support non-AI customers across DDR3/LDDR4/DDR4/KDDR4 and other DRAM products, while prioritizing higher-value applications where supply remains tight.

Q. Given that DDR5 demand is strong, why is DDR5 only around 10% of total revenue?

A. Demand is strong across DDR4, LPDDR4, DDR3, and DDR5, but capacity is limited. DDR3/DDR4 supply is more constrained, and economics remain attractive, so NYT continues to allocate meaningful capacity to these products rather than shifting more aggressively to DDR5.

Q. What applications are driving demand?

A. Consumer-related applications include auto, industrial, and communications. These applications account for roughly 60% of revenue, with the remainder coming from cloud, PC, and mobile.

Q. What is the shareholder return outlook?

A. Dividend payouts will be balanced against capital needs for capex and equipment. There is still room for shareholder returns, with a potential dividend payout of around 40% of net profit.

Q. Were employee bonuses expenses included in 2Q results?

A. Employee bonus expense was included in 1H26 results.

Q. What are the details on the timing of capex spending?

A. Capex cash outflows will follow construction and equipment payment schedules. Payments are generally made after inspection and acceptance, so actual cash spending will lag committed capex.

Q. Is the NT\$52bn 2026 capex plan based on commitments rather than cash payments?

A. Yes. Only NT\$6.9bn was paid in 1H26, and cash payments should increase in coming quarters as projects progress.

Q. Could NYT receive stronger government support?

A. Government support remains fragmented across central, local, and departmental levels, but NYT continues to seek more unified support for Taiwan's DRAM industry.

Q. How does Taiwan's foundry ecosystem help wafer-to-wafer product development?

A. Taiwan's advanced semiconductor ecosystem should help streamline customized memory projects. Several projects are pending, and proximity to leading foundry/packaging capabilities may support wafer-to-wafer development.

Q. Would NYT consider LTA structures such as cash prepayment/price floors/take-or-pay to reduce default risk during downturns?

A. Yes, such structures will be considered.

Q. Given the cyclical nature of memory, what is the outlook from here?

A. Memory remains cyclical, especially for lower-value non-AI products. However, the overall market remains in shortage, supported by AI/server demand and tight capacity across applications.

Q. Comment on NYT LPDDR potentially entering NVIDIA's Vera Rubin supply chain?

A. No comment due to customer confidentiality.

Q. What is the impact of the recent wafer price increases on gross margin?

A. The focus remains on ASP momentum. 2Q ASP increases are expected to continue into 3Q, which should help offset cost pressures.

Q. Is there better pull-in momentum from customers after earlier consumer price resistance?

A. Supply remains tight, limiting the ability to support incremental pull-in momentum.

Investment Thesis, Valuation and Risks

Nanya Technology *(Overweight; Price Target: NT\$710.00)*

Investment Thesis

We are OW on NYT, given the resilient legacy DRAM pricing coupled with leading memory makers phasing out DDR4 to drive pricing leverage higher in 2H26E and 2027E. NYT's entry into the NVIDIA supply chain will help it to extend its server industry exposure well beyond the 20% level, while simultaneously reducing its large CE exposure, making its pricing trend less volatile and more stable. While we expect quarterly earnings y-y momentum to slightly decelerate as we head into 2H26, we see NYT as having a relatively favorable risk/reward in the near-mid term as we expect all memory stocks to move up together during an up-cycle.

Valuation

Our June-27 PT of NT\$710 is based on 8x FY27E EPS, on par with the Korean DRAM makers driven by our higher-for-longer thesis (vs. previously diverging pricing momentum expectations between DDR4 and DDR5).

Risks to Rating and Price Target

Downside risks include:

- Shorter-than-expected legacy DRAM spot price momentum into 2H26.
- A shorter-than-expected D4 16Gb die shelf cycle from the consumer market (i.e., customers' willingness to transition from D4 to D5, could result in weaker than expected D4/LPD4 pricing from 2026-end to 2027 onwards).
- Unsuccessful AI-related memory product development, allowing the company to de-rate (a failure in next-gen technology migration, creating room for competitors to catch up).
- Faster-than-expected DDR5 yield improvement from CXMT.

Analyst Certification: The Research Analyst(s) denoted by an “AC” on the cover of this report certifies (or, where multiple Research Analysts are primarily responsible for this report, the Research Analyst denoted by an “AC” on the cover or within the document individually certifies, with respect to each security or issuer that the Research Analyst covers in this research) that: (1) all of the views expressed in this report accurately reflect the Research Analyst’s personal views about any and all of the subject securities or issuers; and (2) no part of any of the Research Analyst’s compensation was, is, or will be directly or indirectly related to the specific recommendations or views expressed by the Research Analyst(s) in this report. For all Korea-based Research Analysts listed on the front cover, if applicable, they also certify, as per KOFIA requirements, that the Research Analyst’s analysis was made in good faith and that the views reflect the Research Analyst’s own opinion, without undue influence or intervention.

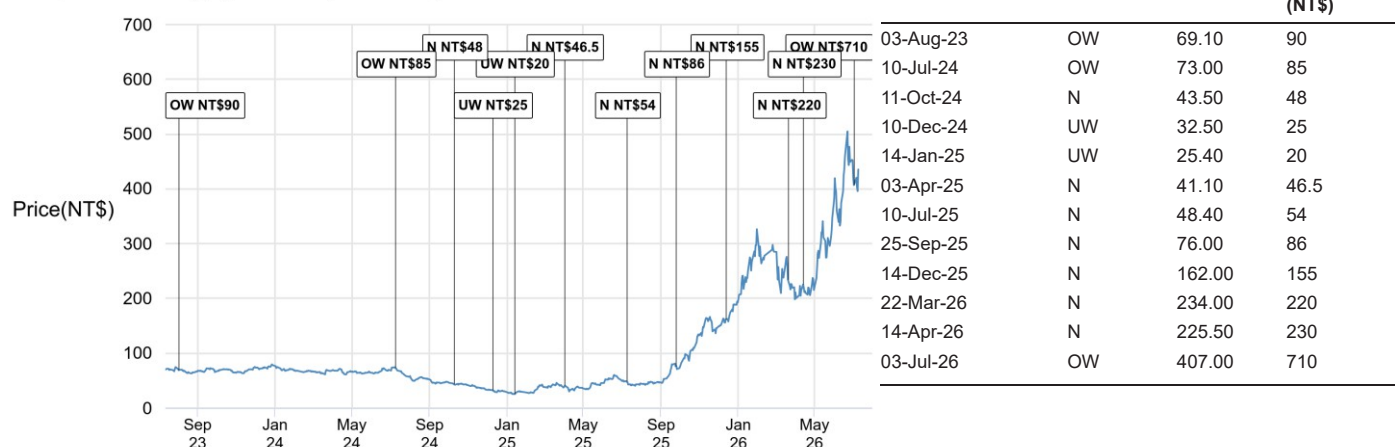
All authors named within this report are Research Analysts who produce independent research unless otherwise specified. In Europe, Sector Specialists (Sales and Trading) may be shown on this report as contacts but are not authors of the report or part of the Research Department.

Important Disclosures

- **Market Maker/ Liquidity Provider:** J.P. Morgan is a market maker and/or liquidity provider in the financial instruments of/related to Nanya Technology or related entities.
- **Client:** J.P. Morgan currently has, or had within the past 12 months, the following entity(ies) as clients: Nanya Technology or related entities.
- **Debt Position:** J.P. Morgan may hold a position in the debt securities of Nanya Technology or related entities, if any.

Company-Specific Disclosures: J.P. Morgan does and seeks to do business with companies covered in its research reports. As a result, investors should be aware that the firm may have a conflict of interest that could affect the objectivity of this report. Investors should consider this report as only a single factor in making their investment decision. Important disclosures, including price charts and credit opinion history tables (if applicable), are available for compendium reports and all J.P. Morgan-covered companies, and certain non-covered companies, by visiting <https://www.jpmm.com/research/disclosures>, calling 1-800-477-0406, or e-mailing research.disclosure.inquiries@jpmorgan.com with your request.

Nanya Technology (2408.TW, 2408 TT) Price Chart



Source: Bloomberg Finance L.P. and J.P. Morgan; price data adjusted for stock splits and dividends.
Initiated coverage Jun 11, 2018. All share prices are as of market close on the previous business day.

The chart(s) show J.P. Morgan's continuing coverage of the stocks; the current analysts may or may not have covered it over the entire period. J.P. Morgan ratings or designations: OW = Overweight, N= Neutral, UW = Underweight, NR = Not Rated

Explanation of Equity Research Ratings, Designations and Analyst(s) Coverage Universe:

J.P. Morgan uses the following rating system: Overweight (over the duration of the price target indicated in this report, we expect this stock will outperform the average total return of the stocks in the Research Analyst’s, or the Research Analyst’s team’s, coverage universe); Neutral (over the duration of the price target indicated in this report, we expect this stock will perform in line with the average total return of the stocks in the Research Analyst’s, or the Research Analyst’s team’s, coverage universe); and Underweight (over the duration of the price target indicated in this report, we expect this stock will underperform the average total return of the stocks in the Research Analyst’s, or the Research Analyst’s team’s, coverage universe. NR is Not Rated. In this case, J.P. Morgan has removed the rating and, if applicable, the price target, for this stock because of either a lack of a sufficient fundamental basis or for legal, regulatory or policy reasons. The previous rating and, if applicable, the price target, no longer should be relied upon. An NR designation is not a recommendation or a rating. Some stocks under coverage have a rating but no price target; in these cases, we expect the stock will outperform/perform in line/underperform the average total return of the stocks in the

Research Analyst's, or the Research Analyst's team's, coverage universe of the relevant duration of the region. In our Asia (ex-Australia and ex-India) and U.K. small- and mid-cap Equity Research, each stock's expected total return is compared to the expected total return of a benchmark country market index, not to those Research Analysts' coverage universe. If it does not appear in the Important Disclosures section of this report, the certifying Research Analyst's coverage universe can be found on J.P. Morgan's Research website, <https://www.jpmorganmarkets.com>.

Coverage Universe: Kwon, Jay : LG Corp (003550.KS), LG Display (034220.KS), LG Electronics (066570.KS), LG Innotek (011070.KS), Nanya Technology (2408.TW), SK Inc (034730.KS), SK hynix (000660.KS), SKC (011790.KS), Samsung C&T (028260.KS), Samsung Electro-Mechanics (009150.KS), Samsung Electronics (005930.KS), Samsung SDS (018260.KS)

J.P. Morgan Equity Research Ratings Distribution, as of July 04, 2026

	Overweight (buy)	Neutral (hold)	Underweight (sell)
J.P. Morgan Global Equity Research Coverage*	53%	36%	12%
IB clients**	83%	80%	73%
JPMS Equity Research Coverage*	51%	37%	12%
IB clients**	95%	92%	87%

*Please note that the percentages may not add to 100% because of rounding.

**Percentage of subject companies within each of the "buy," "hold" and "sell" categories for which J.P. Morgan has provided investment banking services within the previous 12 months.

For purposes of FINRA ratings distribution rules only, our Overweight rating falls into a buy rating category; our Neutral rating falls into a hold rating category; and our Underweight rating falls into a sell rating category. Please note that stocks with an NR designation are not included in the table above. This information is current as of the end of the most recent calendar quarter.

Equity Valuation and Risks: For valuation methodology and risks associated with covered companies or price targets for covered companies, please see the most recent company-specific research report at <http://www.jpmorganmarkets.com>, contact the primary analyst or your J.P. Morgan representative, or email research.disclosure.inquiries@jpmorgan.com. For material information about the proprietary models used, please see the Summary of Financials in company-specific research reports and the Company Tearsheets, which are available to download on the company pages of our client website, <http://www.jpmorganmarkets.com>. This report also sets out within it the material underlying assumptions used.

History of Investment Recommendations:

A history of J.P. Morgan investment recommendations disseminated during the preceding 12 months can be accessed on the Research & Commentary page of <http://www.jpmorganmarkets.com> where you can also search by analyst name, sector or financial instrument.

Analysts' Compensation: The research analysts responsible for the preparation of this report receive compensation based upon various factors, including the quality and accuracy of research, client feedback, competitive factors, and overall firm revenues.

Registration of non-US Analysts: Unless otherwise noted, the non-US analysts listed on the front of this report are employees of non-US affiliates of J.P. Morgan Securities LLC, may not be registered as research analysts under FINRA rules, may not be associated persons of J.P. Morgan Securities LLC, and may not be subject to FINRA Rule 2241 or 2242 restrictions on communications with covered companies, public appearances, and trading securities held by a research analyst account.

Other Disclosures

J.P. Morgan is a marketing name for investment banking businesses of JPMorgan Chase & Co. and its subsidiaries and affiliates worldwide.

UK MIFID FICC research unbundling exemption: UK clients should refer to [UK MIFID Research Unbundling exemption](#) for details of J.P. Morgan's implementation of the FICC research exemption and guidance on relevant FICC research categorisation.

All research material made available to clients are simultaneously available on our client website, J.P. Morgan Markets, unless specifically permitted by relevant laws. Not all research content is redistributed, e-mailed or made available to third-party aggregators. For all research material available on a particular stock, please contact your sales representative.

Any long form nomenclature for references to China; Hong Kong; Taiwan; and Macau within this research material are Mainland China; Hong Kong SAR (China); Taiwan (China); and Macau SAR (China).

J.P. Morgan Research may, from time to time, write on issuers or securities targeted by economic or financial sanctions imposed or administered by the governmental authorities of the U.S., EU, UK or other relevant jurisdictions (Sanctioned Securities). Nothing in this report is intended to be read or construed as encouraging, facilitating, promoting or otherwise approving investment or dealing in such Sanctioned Securities. Clients should be aware of their own legal and compliance obligations when making investment decisions.

Any digital or crypto assets discussed in this research report are subject to a rapidly changing regulatory landscape. For relevant regulatory advisories on crypto assets, including bitcoin and ether, please see <https://www.jpmorgan.com/disclosures/cryptoasset-disclosure>.

The author(s) of this research report may not be licensed to carry on regulated activities in your jurisdiction and, if not licensed, do not hold themselves out as being able to do so.

Exchange-Traded Funds (ETFs): J.P. Morgan Securities LLC ("JPMS") acts as authorized participant for substantially all U.S.-listed ETFs. To the extent that any ETFs are mentioned in this report, JPMS may earn commissions and transaction-based compensation in connection with the distribution of those ETF shares and may earn fees for performing other trade-related services, such as securities lending to short sellers of the ETF shares. JPMS may also perform services for the ETFs themselves, including acting as a broker or dealer to the ETFs. In addition, affiliates of JPMS may perform services for the ETFs, including trust, custodial, administration, lending, index calculation and/or maintenance and other services.

Options and Futures related research: If the information contained herein regards options- or futures-related research, such information is available only to persons who have received the proper options or futures risk disclosure documents. Please contact your J.P. Morgan Representative or visit <https://www.theocc.com/components/docs/riskstoc.pdf> for a copy of the Option Clearing Corporation's Characteristics and Risks of Standardized Options or https://www.finra.org/sites/default/files/2020-08/Security_Futures_Risk_Disclosure_Statement_2020.pdf for a copy of the Security Futures Risk Disclosure Statement.

Changes to Interbank Offered Rates (IBORs) and other benchmark rates: Certain interest rate benchmarks are, or may in the future become, subject to ongoing international, national and other regulatory guidance, reform and proposals for reform. For more information, please consult: https://www.jpmorgan.com/global/disclosures/interbank_offered_rates

Private Bank Clients: Where you are receiving research as a client of the private banking businesses offered by JPMorgan Chase & Co. and its subsidiaries ("J.P. Morgan Private Bank"), research is provided to you by J.P. Morgan Private Bank and not by any other division of J.P. Morgan, including, but not limited to, the J.P. Morgan Corporate and Investment Bank and its Global Research division.

Legal entity responsible for the production and distribution of research: The legal entity identified below the name of the Reg AC Research Analyst who authored this material is the legal entity responsible for the production of this research. Where multiple Reg AC Research Analysts authored this material with different legal entities identified below their names, these legal entities are jointly responsible for the production of this research. Where more than one legal entity is listed under an analyst's name, the first legal entity is responsible for the production unless stated otherwise. Research Analysts from various J.P. Morgan affiliates may have contributed to the production of this material but may not be licensed to carry out regulated activities in your jurisdiction (and do not hold themselves out as being able to do so). Unless otherwise stated below in the legal entity disclosures, this material has been distributed by the legal entity responsible for production, or where more than one legal entity is listed under the analyst's name, the first legal entity will be responsible for distribution. If you have any queries, please contact the relevant Research Analyst in your jurisdiction or the entity in your jurisdiction that has distributed this research material.

Legal Entities Disclosures and Country-/Region-Specific Disclosures:

Argentina: JPMorgan Chase Bank N.A Sucursal Buenos Aires is regulated by Banco Central de la República Argentina ("BCRA"- Central Bank of Argentina) and Comisión Nacional de Valores ("CNV"- Argentinian Securities Commission - ALYC y AN Integral N°51).

Australia: J.P. Morgan Securities Australia Limited ("JPMSAL") (ABN 61 003 245 234/AFS Licence No: 238066) is regulated by the Australian Securities and Investments Commission and is a Market Participant of ASX Limited, a Clearing and Settlement Participant of ASX Clear Pty Limited and a Clearing Participant of ASX Clear (Futures) Pty Limited. This material is issued and distributed in Australia by or on behalf of JPMSAL only to "wholesale clients" (as defined in section 761G of the Corporations Act 2001). A list of all financial products covered can be found by visiting <https://www.jpmm.com/research/disclosures>. J.P. Morgan seeks to cover companies of relevance to the domestic and international investor base across all Global Industry Classification Standard (GICS) sectors, as well as across a range of market capitalisation sizes. If applicable, in the course of conducting public side due diligence on the subject company(ies), the Research Analyst team may at times perform such diligence through corporate engagements such as site visits, discussions with company representatives, management presentations, etc. Research issued by JPMSAL has been prepared in accordance with J.P. Morgan Australia's Research Independence Policy which can be found at the following link: [J.P. Morgan Australia - Research Independence Policy](#).

Brazil: Banco J.P. Morgan S.A. is regulated by the Comissão de Valores Mobiliários (CVM) and by the Central Bank of Brazil. Ombudsman J.P. Morgan: 0800-7700847 / 0800-7700810 (For Hearing Impaired) / ouvidoria.jp.morgan@jpmchase.com.

Canada: J.P. Morgan Securities Canada Inc. is a registered investment dealer, regulated by the Canadian Investment Regulatory Organization and the Ontario Securities Commission and is the participating member on Canadian exchanges. This material is distributed in Canada by or on behalf of J.P.Morgan Securities Canada Inc.

Chile: Inversiones J.P. Morgan Limitada is an unregulated entity incorporated in Chile.

China: J.P. Morgan Securities (China) Company Limited has been approved by CSRC to conduct the securities investment consultancy business.

Colombia: Banco J.P. Morgan Colombia S.A. is supervised by the Superintendencia Financiera de Colombia (SFC). Any reference in this material to products or services offered abroad by entities other than the Bank in Colombia is included exclusively for descriptive purposes. Such references do not constitute, and should not be construed as, promotional activity or the provision of financial products or services within Colombian territory, as defined under applicable Colombian regulation.

Dubai International Financial Centre (DIFC): JPMorgan Chase Bank, N.A., Dubai Branch is regulated by the Dubai Financial Services Authority (DFSA) and its registered address is Dubai International Financial Centre - The Gate, West Wing, Level 3 and 9 PO Box 506551, Dubai, UAE. This material has been distributed by JP Morgan Chase Bank, N.A., Dubai Branch to persons regarded as professional clients or market counterparties as defined under the DFSA rules.

European Economic Area (EEA): Unless specified to the contrary, research is distributed in the EEA by J.P. Morgan SE ("JPM SE"), which is authorised as a credit institution by the Federal Financial Supervisory Authority (Bundesanstalt für Finanzdienstleistungsaufsicht, BaFin) and jointly supervised by the BaFin, the German Central Bank (Deutsche Bundesbank) and the European Central Bank (ECB). JPM SE is a company headquartered in Frankfurt with registered address at TaunusTurm, Taunustor 1, Frankfurt am Main, 60310, Germany. The material has been distributed in the EEA to persons regarded as professional investors (or equivalent) pursuant to Art. 4 para. 1 no. 10 and Annex II of MiFID II and its respective implementation in their home jurisdictions ("EEA professional investors"). This material must not be acted on or relied on by persons who are not EEA professional investors. Any investment or investment activity to which this material relates is only available to EEA relevant persons and will be engaged in only with EEA relevant persons.

Hong Kong: J.P. Morgan Securities (Asia Pacific) Limited (CE number AAJ321) is regulated by the Hong Kong Monetary Authority and the Securities and Futures Commission in Hong Kong, and J.P. Morgan Broking (Hong Kong) Limited (CE number AAB027) is regulated by the Securities and Futures Commission in Hong Kong. JP Morgan Chase Bank, N.A., Hong Kong Branch (CE Number AAL996) is regulated by the Hong Kong Monetary Authority and the Securities and Futures Commission, is organized under the laws of the United States with limited liability. Where the distribution of this material is a regulated activity in Hong Kong, the material is distributed in Hong Kong by or through J.P. Morgan Securities (Asia Pacific) Limited and/or J.P. Morgan Broking (Hong Kong) Limited.

India: J.P. Morgan India Private Limited (Corporate Identity Number - U67120MH1992FTC068724), having its registered office at J.P. Morgan Tower, Off. C.S.T. Road, Kalina, Santacruz - East, Mumbai – 400098, is registered with the Securities and Exchange Board of India (SEBI) as a 'Research Analyst' having registration number INH000001873. J.P. Morgan India Private Limited is also registered with SEBI as a member of the National Stock Exchange of India Limited and the Bombay Stock Exchange Limited (SEBI Registration Number – INZ000239730) and as a Merchant Banker (SEBI Registration Number - MB/INM000002970). Telephone: 91-22-6157 3000, Facsimile: 91-22-6157 3990 and Website: <http://www.jpmpil.com>. JPMorgan Chase Bank, N.A. - Mumbai Branch is licensed by the Reserve Bank of India (RBI) (Licence No. 53/ Licence No. BY.4/94; SEBI - IN/CUS/014/ CDSL : IN-DP-CDSL-444-2008/ IN-DP-NSDL-285-2008/ INBI00000984/ INE231311239) as a Scheduled Commercial Bank in India, which is its primary license allowing it to carry on Banking business in India and other activities, which a Bank branch in India are permitted to undertake. For non-local research material, this material is not distributed in India by J.P. Morgan India Private Limited. Compliance Officer: Ashutosh Sharma; ashutosh.j.sharma@jpmchase.com; +912261575002. Grievance Officer: Ramprasad K, jpmpil.research.feedback@jpmorgan.com; +912261573000. Registration granted by SEBI and certification from NISM in no way guarantee performance of the intermediary or provide any assurance of returns to investors. Please visit [Terms and Conditions and Most Important Terms and Conditions \(MITC\)](#). The annual Compliance audit report is available at <http://www.jpmpil.com/#research>.

Indonesia: PT J.P. Morgan Sekuritas Indonesia is a member of the Indonesia Stock Exchange and is registered and supervised by the Otoritas Jasa Keuangan (OJK).

Korea: J.P. Morgan Securities (Far East) Limited, Seoul Branch, is a member of the Korea Exchange (KRX). JPMorgan Chase Bank, N.A., Seoul Branch, is licensed as a branch office of foreign bank (JPMorgan Chase Bank, N.A.) in Korea. Both entities are regulated by the Financial Services Commission (FSC) and the Financial Supervisory Service (FSS). For non-macro research material, the material is distributed in Korea by or through J.P. Morgan Securities (Far East) Limited, Seoul Branch.

Japan: JPMorgan Securities Japan Co., Ltd. and JPMorgan Chase Bank, N.A., Tokyo Branch are regulated by the Financial Services Agency in Japan.

Malaysia: This material is issued and distributed in Malaysia by JPMorgan Securities (Malaysia) Sdn Bhd (18146-X), which is a Participating Organization of Bursa Malaysia Berhad and holds a Capital Markets Services License issued by the Securities Commission in Malaysia.

Mexico: J.P. Morgan Casa de Bolsa, S.A. de C.V., J.P. Morgan Grupo Financiero is member of the Mexican Stock Exchange ("Bolsa Mexicana de Valores") and the Institutional Stock Exchange ("Bolsa Institucional de Valores"), and it is authorized to act as a broker dealer by the National Banking and Securities Exchange Commission ("Comisión Nacional Bancaria y de Valores").

New Zealand: This material is issued and distributed by JPMSAL in New Zealand only to "wholesale clients" (as defined in the Financial Markets Conduct Act 2013). JPMSAL is registered as a Financial Service Provider under the Financial Service providers (Registration and Dispute Resolution) Act of 2008.

Philippines: J.P. Morgan Securities Philippines Inc. is a Trading Participant of the Philippine Stock Exchange and a member of the Securities Clearing Corporation of the Philippines and the Securities Investor Protection Fund. It is regulated by the Securities and Exchange Commission.

Singapore: This material is issued and distributed in Singapore by or through J.P. Morgan Securities Singapore Private Limited (JPMS) [MDDI (P) 057/08/2025 and Co. Reg. No.: 199405335R], which is a member of the Singapore Exchange Securities Trading Limited, and/or JPMorgan Chase Bank, N.A., Singapore branch (JPMCB Singapore), both of which are regulated by the Monetary Authority of Singapore. This material is issued and distributed in Singapore only to accredited investors, expert investors and institutional investors, as defined in Section 4A of the Securities and Futures Act, Cap. 289 (SFA). This material is not intended to be issued or distributed to any retail investors or any other investors that do not fall into the classes of "accredited investors," "expert investors" or "institutional investors," as defined under Section 4A of

the SFA. Recipients of this material in Singapore are to contact JPMSS or JPMCB Singapore in respect of any matters arising from, or in connection with, the material.

South Africa: J.P. Morgan Equities South Africa Proprietary Limited and JPMorgan Chase Bank, N.A., Johannesburg Branch are members of the Johannesburg Securities Exchange and are regulated by the Financial Services Conduct Authority (FSCA).

Taiwan: J.P. Morgan Securities (Taiwan) Limited is a participant of the Taiwan Stock Exchange (company-type) and regulated by the Taiwan Securities and Futures Bureau. Material relating to equity securities is issued and distributed in Taiwan by J.P. Morgan Securities (Taiwan) Limited, subject to the license scope and the applicable laws and the regulations in Taiwan. **To the extent that J.P. Morgan Securities (Taiwan) Limited produces research materials on securities not listed on the Taiwan Stock Exchange or Taipei Exchange (“Non-Taiwan Listed Securities”), these materials shall not constitute securities recommendations for the purpose of applicable Taiwan regulations, and, for the avoidance of doubt, J.P. Morgan Securities (Taiwan) Limited does not act as broker for Non-Taiwan Listed Securities.** According to Paragraph 2, Article 7-1 of Operational Regulations Governing Securities Firms Recommending Trades in Securities to Customers (as amended or supplemented) and/or other applicable laws or regulations, please note that the recipient of this material is not permitted to engage in any activities in connection with the material that may give rise to conflicts of interests, unless otherwise disclosed in the “Important Disclosures” in this material.

Thailand: This material is issued and distributed in Thailand by JPMorgan Securities (Thailand) Ltd., which is a member of the Stock Exchange of Thailand and is regulated by the Ministry of Finance and the Securities and Exchange Commission. The registered address is 548 One City Center Building, 50th Floor, Ploenchit Road, Lumpini, Pathum Wan, Bangkok 10330.

UK: Research is produced in the UK by J.P. Morgan Securities plc (“JPMS plc”) which is a member of the London Stock Exchange and is authorised by the Prudential Regulation Authority and regulated by the Financial Conduct Authority and the Prudential Regulation Authority or J.P. Morgan Markets Limited (“JPMML Ltd”) which is authorised and regulated by the Financial Conduct Authority. Unless specified to the contrary, this material is distributed in the UK by JPMS plc and is directed in the UK only to: (a) persons having professional experience in matters relating to investments falling within article 19(5) of the Financial Services and Markets Act 2000 (Financial Promotion) (Order) 2005 (“the FPO”); (b) persons outlined in article 49 of the FPO (high net worth companies, unincorporated associations or partnerships, the trustees of high value trusts, etc.); or (c) any persons to whom this communication may otherwise lawfully be made; all such persons being referred to as “UK relevant persons”. This material must not be acted on or relied on by persons who are not UK relevant persons. Any investment or investment activity to which this material relates is only available to UK relevant persons and will be engaged in only with UK relevant persons. A description of J.P. Morgan EMEA’s policy for prevention and avoidance of conflicts of interest related to the production of Research can be found at the following link: [J.P. Morgan EMEA - Research Independence Policy](#).

U.S.: J.P. Morgan Securities LLC (“JPMS”) is a member of the NYSE, FINRA, SIPC, and the NFA. JPMorgan Chase Bank, N.A. is a member of the FDIC. Material published by non-U.S. affiliates is distributed in the U.S. by JPMS who accepts responsibility for its content.

General: Additional information is available upon request. The information in this material has been obtained from sources believed to be reliable. While all reasonable care has been taken to ensure that the facts stated in this material are accurate and that the forecasts, opinions and expectations contained herein are fair and reasonable, JPMorgan Chase & Co. or its affiliates and/or subsidiaries (collectively J.P. Morgan) make no representations or warranties whatsoever to the completeness or accuracy of the material provided, except with respect to any disclosures relative to J.P. Morgan and the Research Analyst's involvement with the issuer that is the subject of the material. Accordingly, no reliance should be placed on the accuracy, fairness or completeness of the information contained in this material. There may be certain discrepancies with data and/or limited content in this material as a result of calculations, adjustments, translations to different languages, and/or local regulatory restrictions, as applicable. These discrepancies should not impact the overall investment analysis, views and/or recommendations of the subject company(ies) that may be discussed in the material. Artificial intelligence tools may have been used in the preparation of this material, including assisting in data analysis, pattern recognition, and content drafting for research material. J.P. Morgan accepts no liability whatsoever for any loss arising from any use of this material or its contents, and neither J.P. Morgan nor any of its respective directors, officers or employees, shall be in any way responsible for the contents hereof, apart from the liabilities and responsibilities that may be imposed on them by the relevant regulatory authority in the jurisdiction in question, or the regulatory regime thereunder. Opinions, forecasts or projections contained in this material represent J.P. Morgan's current opinions or judgment as of the date of the material only and are therefore subject to change without notice. Periodic updates may be provided on companies/industries based on company-specific developments or announcements, market conditions or any other publicly available information. There can be no assurance that future results or events will be consistent with any such opinions, forecasts or projections, which represent only one possible outcome. Furthermore, such opinions, forecasts or projections are subject to certain risks, uncertainties and assumptions that have not been verified, and future actual results or events could differ materially. The value of, or income from, any investments referred to in this material may fluctuate and/or be affected by changes in exchange rates. All pricing is indicative as of the close of market for the securities discussed, unless otherwise stated. Past performance is not indicative of future results. Accordingly, investors may receive back less than originally invested. This material is not intended as an offer or solicitation for the purchase or sale of any financial instrument. The opinions and recommendations herein do not take into account individual client circumstances, objectives, or needs and are not intended as recommendations of particular securities, financial instruments or strategies to particular clients. This material may include views on structured securities, options, futures and other derivatives. These are complex instruments, may involve a high degree of risk and may be appropriate investments only for sophisticated investors who are capable of understanding and assuming the risks involved. The recipients of this material must make their own independent decisions regarding any securities or financial instruments mentioned herein and should seek advice from such independent financial, legal, tax or other adviser as they deem necessary. J.P. Morgan may trade as a principal on the basis of the Research Analysts' views and research, and it may also engage in transactions for its own account or for its clients' accounts in a

manner inconsistent with the views taken in this material, and J.P. Morgan is under no obligation to ensure that such other communication is brought to the attention of any recipient of this material. Others within J.P. Morgan, including Strategists, Sales staff and other Research Analysts, may take views that are inconsistent with those taken in this material. Employees of J.P. Morgan not involved in the preparation of this material may have investments in the securities (or derivatives of such securities) mentioned in this material and may trade them in ways different from those discussed in this material. This material is not an advertisement for or marketing of any issuer, its products or services, or its securities in any jurisdiction.

Confidentiality and Security Notice: This transmission may contain information that is privileged, confidential, legally privileged, and/or exempt from disclosure under applicable law. If you are not the intended recipient, you are hereby notified that any disclosure, copying, distribution, or use of the information contained herein (including any reliance thereon) is STRICTLY PROHIBITED. Although this transmission and any attachments are believed to be free of any virus or other defect that might affect any computer system into which it is received and opened, it is the responsibility of the recipient to ensure that it is virus free and no responsibility is accepted by JPMorgan Chase & Co., its subsidiaries and affiliates, as applicable, for any loss or damage arising in any way from its use. If you received this transmission in error, please immediately contact the sender and destroy the material in its entirety, whether in electronic or hard copy format. This message is subject to electronic monitoring: <https://www.jpmorgan.com/disclosures/email>

MSCI: Certain information herein ("Information") is reproduced by permission of MSCI Inc., its affiliates and information providers ("MSCI") ©2026. No reproduction or dissemination of the Information is permitted without an appropriate license. MSCI MAKES NO EXPRESS OR IMPLIED WARRANTIES (INCLUDING MERCHANTABILITY OR FITNESS) AS TO THE INFORMATION AND DISCLAIMS ALL LIABILITY TO THE EXTENT PERMITTED BY LAW. No Information constitutes investment advice, except for any applicable Information from MSCI ESG Research. Subject also to [msci.com/disclaimer](https://www.msci.com/disclaimer)

Sustainalytics: Certain information, data, analyses and opinions contained herein are reproduced by permission of Sustainalytics and: (1) includes the proprietary information of Sustainalytics; (2) may not be copied or redistributed except as specifically authorized; (3) do not constitute investment advice nor an endorsement of any product or project; (4) are provided solely for informational purposes; and (5) are not warranted to be complete, accurate or timely. Sustainalytics is not responsible for any trading decisions, damages or other losses related to it or its use. The use of the data is subject to conditions available at <https://www.sustainalytics.com/legal-disclaimers>. ©2026 Sustainalytics. All Rights Reserved.

"Other Disclosures" last revised July 04, 2026.

Copyright 2026 JPMorgan Chase & Co. All rights reserved. This material or any portion hereof may not be reprinted, sold or redistributed without the written consent of J.P. Morgan. It is strictly prohibited to use or share without prior written consent from J.P. Morgan any research material received from J.P. Morgan or an authorized third-party ("J.P. Morgan Data") in any third-party artificial intelligence ("AI") systems or models when such J.P. Morgan Data is accessible by a third-party.

Completed 12 Jul 2026 03:13 PM HKT

Disseminated 12 Jul 2026 03:23 PM HKT